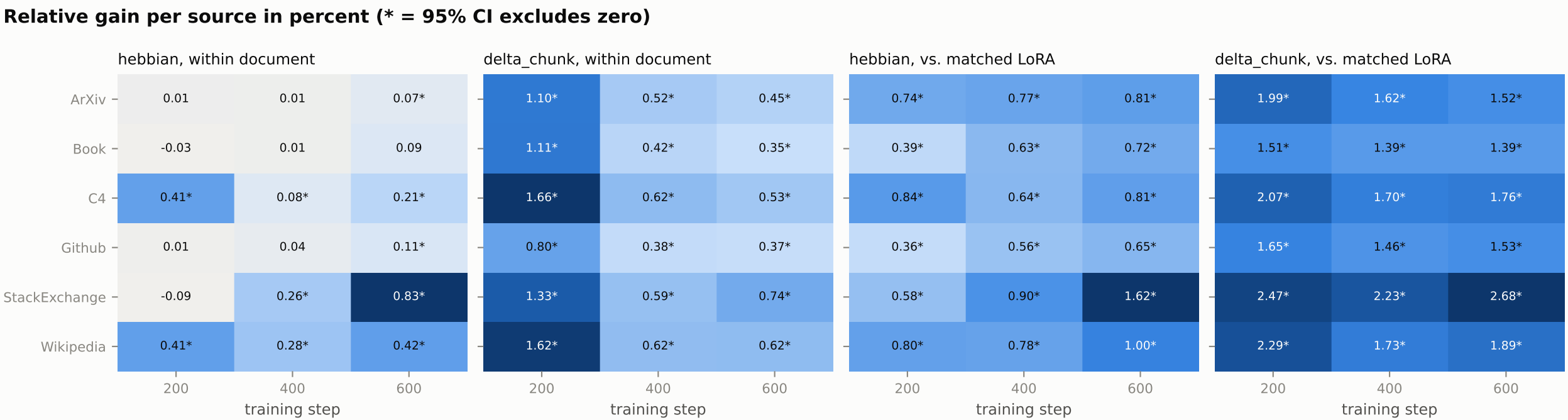
Relative gain per source in percent (* = 95% CI excludes zero)
hebbian, within document
delta_chunk, within document
hebbian, vs. matched LoRA
delta_chunk, vs. matched LoRA
ArXiv
Book
C4
Github
StackExchange
Wikipedia
0.01 0.01 0.07*
-0.03 0.01 0.09
0.41* 0.08* 0.21*
0.01 0.04 0.11*
-0.09 0.26* 0.83*
0.41* 0.28* 0.42*
1.10* 0.52* 0.45*
1.11* 0.42* 0.35*
1.66* 0.62* 0.53*
0.80* 0.38* 0.37*
1.33* 0.59* 0.74*
1.62* 0.62* 0.62*
0.74* 0.77* 0.81*
0.39* 0.63* 0.72*
0.84* 0.64* 0.81*
0.36* 0.56* 0.65*
0.58* 0.90* 1.62*
0.80* 0.78* 1.00*
1.99* 1.62* 1.52*
1.51* 1.39* 1.39*
2.07* 1.70* 1.76*
1.65* 1.46* 1.53*
2.47* 2.23* 2.68*
2.29* 1.73* 1.89*
200 400 600
training step